

Giridhar Pai

Cybersecurity Analyst | SOC Analyst | Blue Team | Digital Forensics

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Professional Summary

Adversarial-minded specialist focused on proactive defense and threat hunting. Expert in vulnerability remediation and detection engineering with proven proficiency in Splunk-driven workflows. Top 1% global performer on TryHackMe, bridging offensive tradecraft with defensive precision to secure complex digital infrastructure. Dedicated to building resilient, automated security architectures.

Core Competencies

- Security monitoring, alert review, and incident analysis.
- Vulnerability assessment and secure application review.
- Digital forensics and metadata-based investigation.
- Documentation, reporting, evidence tracking, and process adherence.
- Communication, teamwork, client-facing professionalism, and coordination.
- Time management, structured follow-up, and task ownership.
- Strong attention to detail, confidentiality, and accountability.

Education

Yenepoya University, Mangalore

Jul 2023 – Jul 2026

BCA in Cyber Security, Ethical Hacking, and Digital Forensics (IBM Industry Collaboration)

CGPA: 9.15

- Specialized in cybersecurity fundamentals, ethical hacking, digital forensics, and practical defense-oriented workflows.

S N Moodbidri Polytechnic

Aug 2019 – May 2022

Diploma in Computer Science and Engineering

Certifications

- **Security Operations & SIEM**
 - [Splunk Training: 10 courses including Enterprise Security, SOAR, and Defense Analyst](#)
 - [Splunk: The Complete Course from Zero to Hero](#)
- **Cloud, Network, and Cybersecurity Foundations**
 - [AWS Certified Security – Specialty](#)
 - [CompTIA Network+](#)
 - [Google Cybersecurity Specialization](#)
 - [Digital Forensics Essentials \(EC-Council\)](#)
- **Practical Simulations and Platforms**
 - [Deloitte Australia Cyber Job Simulation \(Forge\)](#)
 - [SAP Analytics Cloud: Managing Security and Administration](#)
 - [TryHackMe – 13 certifications; Top 1% globally](#)

Technical Skills

- **Security Operations:** SIEM (Splunk), alert review, threat hunting, incident response basics, vulnerability assessment, DDoS mitigation, IDS/IPS concepts.
- **Compliance & Governance:** Documentation support, evidence tracking, process adherence, confidentiality, audit-oriented record keeping.
- **Cloud Security:** AWS Security, IAM, Secrets Manager, secure architecture.
- **Forensics:** Digital forensics, metadata extraction, heatmap analysis.
- **Productivity & Coordination:** Microsoft Excel, Google Sheets, trackers, documentation workflows, and reporting support.

Tools & Technologies

Splunk (Enterprise Security, SOAR), Linux, Docker, Python, Java, C, JavaScript, SQL, HTML/CSS, Git, Excel, Google Sheets, Wireshark, Nmap, Burp Suite, Kali Linux, TryHackMe, Security Onion, ELK Stack, Jira

Experience

IBM Internship – Tejas Raksha (Yenepoya University)

Mar 2026 – Apr 2026

Cybersecurity Intern – Zero Day Scanner for Agricultural Portal – Cybersecurity Developer

- Developed an automated web vulnerability scanner for agriculture portals to identify OWASP Top 10 issues and

support secure application review.

- Contributed to vulnerability detection workflows and secure testing practices relevant to real-world web application security.
- Documented findings clearly and accurately with remediation guidance and secure coding recommendations.
- Maintained structured technical notes and organized outputs to support follow-up, review, and reporting.

IBM Academic Internship / Industry Collaboration (Yenepoya University)

2025 – 2026

Cybersecurity Intern – DDoS Mitigation System Project – Cybersecurity Developer

- Worked on an IBM-linked academic cybersecurity project focused on defensive security, monitoring, and digital forensics as part of a team.
- Built a DDoS mitigation system to detect suspicious traffic patterns and reduce distributed denial-of-service attacks, focusing on backend development.
- Applied defensive security concepts in traffic analysis, project documentation, and workflow execution.
- Supported coordination and reporting while following defined academic and project deadlines.

Simulations & Training

Deloitte Australia Cyber Job Simulation (Forage)

2025

Cyber Security Simulation – SOC Analyst Role

- Completed enterprise cybersecurity tasks involving security monitoring, incident analysis, and risk assessment in a SOC-style capacity.
- Practiced alert review, structured documentation, and response reporting as part of a simulated Blue Team workflow.
- Strengthened communication, analytical thinking, and problem-solving through online client-facing cybersecurity simulation.

AgileTEC In-Plant Training (Android Training)

Apr 2022

Trainee – Android App Development

- Completed a 3-week in-plant training program focused on Android app development and software engineering workflows.
- Gained practical exposure to development processes, project structure, and implementation discipline.

Projects by Type

Hackathon Projects

AI-Powered Deepfake Detection System

2025

CIDECODE Hackathon – Consolation Prize Winner

- Developed a deepfake detection tool using metadata extraction and heatmap analysis for identifying manipulated media.
- Demonstrated applied digital forensics concepts in a team hackathon setting with Santeena and Santhosh.

DefendX

2025

IBM ICE DAY 2025 – 2nd Runner-Up

- Built a real-time threat detection and monitoring system to demonstrate real-time threat monitoring concepts for enterprise security operations.
- Focused on alerting, monitoring, and defender-centric visibility for security teams in a team hackathon project.

Future City Hackathon Project

2025

Smart City Infrastructure Challenge

- Collaborated on a smart city infrastructure project focused on security and resilience.
- Contributed to design and implementation of secure urban systems.

Internship Projects

Tejas Raksha

2026

IBM Internship – Zero Day Scanner for Agricultural Portals

- Developed a security scanner for agriculture portals to identify OWASP Top 10 vulnerabilities and support secure testing.
- Combined vulnerability assessment concepts with reporting-oriented output for practical review use. Developed the project fully independently.

DDoS Mitigation System (IBM Academic Internship)

2025 – 2026

IBM Industry Collaboration – DDoS Mitigation System Project [OFFLINE]

- Built a DDoS mitigation system to detect suspicious traffic patterns and reduce distributed denial-of-service attacks, focusing on backend development.
- Applied defensive security concepts in traffic analysis, project documentation, and workflow execution as part of a

team.

- Supported coordination and reporting while following defined academic and project deadlines.

Freelancing Projects

Manufacturing Vault

Mar 2026 – Apr 2026

Production-Grade File Encryption Platform

- Built a production-grade file encryption platform using AES-256-GCM and Argon2id key derivation with a zero persistence architecture.
- Deployed on Azure using Purdue Model-aligned 4-tier VNet segmentation and Azure WAF v2 for perimeter defense.
- Achieved 100% pass rate on a 45-point security audit covering cryptography, network isolation, and operational logic.

Kinetic Forge

Mar 2026 – Apr 2026

Supply Chain Security Platform

- Implemented mandatory TOTP-based MFA, Google OAuth 2.0 SSO, and role-based access control for supply chain users.
- Encrypted sensitive cargo telemetry, built a real-time shipment map, and created tamper-resistant audit logging.
- Defeated SQLi, XSS, CSRF, and brute-force attacks using logout rules and rate limiting.

Vault Protocol

Mar 2026 – Apr 2026

Financial Transaction Security System

- Designed a dual-authorization workflow for financial transactions with mandatory MFA re-authentication for auditors.
- Implemented idempotency keys using UUIDv7 and Redis TTL to prevent duplicate financial actions.
- Added encrypted PII storage, scrypt-based password hashing, and hash-chained audit logging.

Retail Data Security

Mar 2026 – Apr 2026

Three-Layer Defense Model

- Architected a three-layer defense model combining Azure WAF v2, VNet isolation, and AES-256-GCM encryption.
- Implemented Argon2id-based key derivation, JWT authentication, Redis-cached keys, and encrypted blob storage.
- Validated the system using tamper tests and web attack resistance checks.

Obsidian Pulse

Mar 2026 – Apr 2026

Healthcare Security Platform

- Developed a healthcare security platform with dual-factor identity restoration, encrypted PHI handling, and real-time telemetry.
- Added brute-force resistance through quarantine logic, Redis-backed idempotent actions, IP blacklisting, and audit logging.
- Built for secure and latency-aware healthcare workflows.

EdTech Security Framework

Mar 2026 – Apr 2026

Passwordless Authentication System

- Implemented FIDO2/WebAuthn passkeys, TOTP fallback, and Google OAuth 2.0 SSO with defense-in-depth controls.
- Added restoration logic, login lockouts, IP blacklisting, encrypted secrets, and tamper-resistant audit verification.
- Designed to demonstrate secure passwordless authentication and audit integrity for educational systems.

Kids Wonder Learn (KWL)

2025 – 2026

Browser-Based Educational Platform [OFFLINE]

- Built a browser-based educational platform for children aged 2–13+ with student, parent, teacher, and admin roles.
- Developed interactive learning modules including games, art tools, story content, login workflows, dashboards, progress tracking, and activity logging.
- Used HTML5, CSS3, Vanilla JavaScript, Supabase, and Netlify to design a modular and scalable front-end architecture.
- Planned backend integration with authentication, PostgreSQL schema design, role mapping, visitor logging, and secure access control.
- Added security-oriented design decisions such as hashed visitor tracking, password handling improvements, and audit-style logging.
- Documented deployment flow, data flow, configuration management, and migration path from localStorage to Supabase.

QuartzTech Trading

2025

Security Framework for Financial Trading Platform

- Developed security infrastructure for financial trading platform with multi-layer authentication and encryption.

Personal Projects

Password Strength Checker

2024

Client-Side Password Entropy Tool

- Built a client-side tool to evaluate password entropy and weak patterns and promote stronger password practices.

Doctors Appointment Booking System

2023

Healthcare Workflow System [OFFLINE]

- Developed a workflow-driven healthcare appointment system with authentication, access control, and offline deployment concepts.
- Received Best Project Award for this system.

Achievements

- Ranked in the Top 1% globally on TryHackMe with 13 certifications.
- Secured 2nd Place in the IBM ICE DAY 2025 CTF Competition.
- Finalist in CODE SPRINT Hackathon (Top 7 of 65 teams).
- Won Consolation Prize in CIDECODE Hackathon.
- Received Best Project Award for the Doctors Appointment Booking System.

Activities and Community

- Member, OWASP Yenepoya.
- Active participant in CTFs, hackathons, cybersecurity learning platforms, and student security activities.

Links

- LinkedIn: [linkedin.com/in/giridharpai227](https://www.linkedin.com/in/giridharpai227)
- GitHub: github.com/Giri227
- Portfolio: giri227.github.io/Giridharpai